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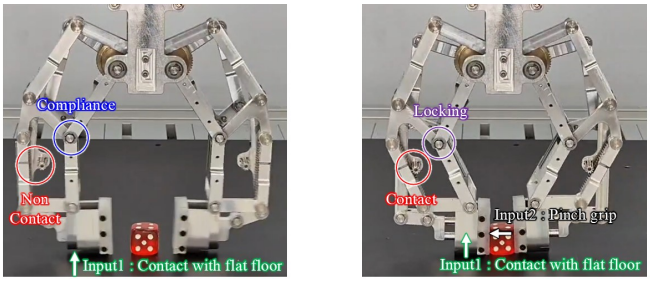
A Novel Under-actuated Gripper based on Passive-Locking Mechanism for Stable Gripping under Environmental Constraints

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***Abstract*— This paper presents a novel under-actuated two-finger gripper that passively adapts to various environments and maintains its grip posture using a passive-locking mechanism. The proposed mechanism features fingers with three phalanges, each incorporating four-bar and eight-bar linkages arranged in parallel. These linkages perform crucial functions, including maintaining the grip angle and ensuring passive characteristics during pinch grips. Previous grippers with passive mechanisms and three-phalanx fingers faced issues with gripping instability, particularly when changes in the passive joint angle were caused by object inertia or external lateral forces. To address this problem, we propose a new passive-locking mechanism utilizing an eight-bar linkage. This innovative design is engineered to adapt to environmental conditions, establish a secure grip, and maintain the grip angle of the passive joint after the grip is achieved. To demonstrate the advantages of the proposed mechanism, this paper conducts a fingertip force vector analysis and a mobility analysis according to the pinch sequence. It also details the derivation process and principles of the mechanism. The gripper's operational range and gripping force are examined through kinematic analysis and verified by simulation. Furthermore, the study shows that the proposed mechanism effectively responds to environmental constraints, even in environments with obstacles surrounding the object. Comparative experiments with and without a contact bar indicate that the proposed gripper can stably secure an object in scenarios involving swing motions and external forces of approximately $5N$.**

I. INTRODUCTION

Robot hands are used to handle various items and operate in frequently changing environments in industrial settings. In such dynamic environments, environmental constraints, such as unknown obstacles, often arise. Consequently, robot hands must exhibit exceptional manual dexterity to navigate these constraints effectively. While incorporating multiple active joints into the fingers can enhance a robot's manual dexterity, this approach also increases the number of actuators and components needed. As a result, the gripper's weight grows, and more complex control strategies are required for the actuators [1]. As an alternative, various studies have explored under-actuated grippers with fewer actuators than the degrees of freedom of finger joints, offering simpler structures and control mechanisms [2-6]. Among these under-actuated grippers, linkage-driven designs feature fingers with two or three phalanges, including both actuating and passive joints. Notable examples include RobotiQ's 2F-140 [7] and DH Robotics' AG-105-145 [8]. These grippers utilize a finger structure comprising four-bar linkages and support a range of gripping methods. The gripping techniques can be classified into two categories: power grasp, which involves wrapping around an object and pulling it inward, and fingertip grip, which involves picking up an object using the gripper's fingertips. Furthermore, the fingertip grip can be further divided into the force-fit grip, which lifts an object vertically, and the form-fit grip, which secures an object in a horizontal position.



(a) (b)

Fig. 1. Pinch sequence of passive locking gripper. (a) responding to flat floor, (b) grip an object with passive-locking

Despite their versatility in various gripping methods, these grippers face structural limitations when dealing with environmental constraints, such as grasping thin objects placed on the floor or objects surrounded by obstacles, due to the limited number of finger phalanges. Research has explored several control methods to address these challenges, including manipulating the inclination of the gripper's fingertip to grasp thin objects placed on the floor [9], implementing force compatibility control for environmental contact [10], and using coil springs to adapt to environmental conditions [11]. However, these solutions are often constrained to specific scenarios or require specialized control techniques.

Some studies [12-13] have introduced two-finger grippers that can adapt to environmental constraints without precise control. These grippers feature finger structures with three phalanges, incorporating four-bar and five-bar linkages arranged in parallel. The four-bar linkage includes a passive joint made of springs, enabling environmental adaptation, while the five-bar linkage ensures parallelism by aligning the fingertips consistently. These grippers are capable of handling objects on the floor or near obstacles without special

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manipulation and are particularly suited for peg-in-hole tasks. However, increasing the number of phalanges without adding actuators makes it challenging to control the passive spring joint while gripping an object. This limitation impedes the ability to maintain the grip posture in 'Form fit grip' mode, where the robot must lift an object horizontally, as it cannot respond effectively to the object's inertia or external forces and is restricted to lifting only light objects [13].

To address this issue, locking the passive joint during gripping could provide a solution. Reference [14] offers an overview of various locking methods and a flowchart for selecting an appropriate method. According to the flowchart, hydraulic, ratchet, latch, or dog clutch methods can be employed for locking in three-phalanx fingers, as they do not require adjustments to locking torque or load-based locking/unlocking. However, each locking method has its limitations: 1) The hydraulic method allows locking/unlocking based on load detection at any position but requires control of the servo valve for these operations; 2) The ratchet method offers multiple locking positions, but controlling an active pawl or using a passive ratchet necessitates a special operational step for effective locking; and 3) The latch or dog clutch methods are limited by the fact that locking is possible only at specific positions due to restricted locking positions. Additionally, various locking systems such as self-amplifying brakes and overrunning clutches are applied to grippers. Since these systems use separate electrical drives, like brakes and motors, they are unsuitable for under-actuated grippers with simple structures and control. Differential gear-based brakes [15] and locking methods that rely on environmental contact [16] achieve passive locking without separate actuators but often require a large number of parts or use soft materials, which makes them impractical for the proposed finger structure that relies solely on rigid links for stability. Consequently, a new linkage-driven locking mechanism is needed to securely grip objects while adapting to diverse objects and unknown environments.

We propose an eight-bar linkage mechanism that enables a gripper with three-phalanx fingers to perform passive locking during a pinch grip, as illustrated in Fig. 1. This mechanism ensures a stable grip by fixing the grip pose even under the object's inertia or external forces acting vertically or laterally on the finger.

The paper is organized as follows: Section II describes the locking principle of the proposed mechanism and verifies its validity through mobility analysis. Section III derives the gripper's working range and the relationship between input torque and fingertip force through kinematic analysis, with formulas and operations verified via simulation. Section IV presents the verification of the gripper's functionality through various tests. Finally, Section V offers conclusions and outlines future work.

II. Finger with Passive-Locking Mechanism

The four-bar linkage mechanism serves as the fundamental component in most link-based under-actuated finger structures. Fig. 2 illustrates the proposed finger structure, which

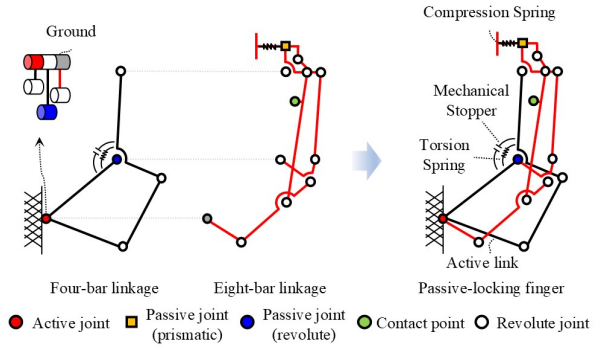


Fig. 2. Schematic diagram of the proposed three-phalanx finger mechanism.

consists of three phalanges with stacked and parallel eight-bar linkages, built upon the basic four-bar linkage design. The four-bar linkage responds to environmental contact through a passive joint that incorporates a torsion spring and a mechanical stopper. The eight-bar linkage maintains the parallel alignment of both fingers' fingertips by forming a double parallelogram. Additionally, a prismatic joint, equipped with a compression spring near the fingertips, plays a crucial role in implementing locking and unlocking during pinch grip situations. This mechanism is based on the locking principle derived from the fingertip configuration and the force vector characteristics in the four-bar linkage. The validity of the derived mechanism is confirmed through mobility analysis.

A. Descriptions of the locking grip for different situations

Fig. 3 illustrates the difference in gripper joint operation depending on the presence of the locking mechanism across two pinch grip scenarios. In all two scenarios, without the locking mechanism, the passive joints of the fingers rotate due to the fingertip force directed toward the gripper's palm

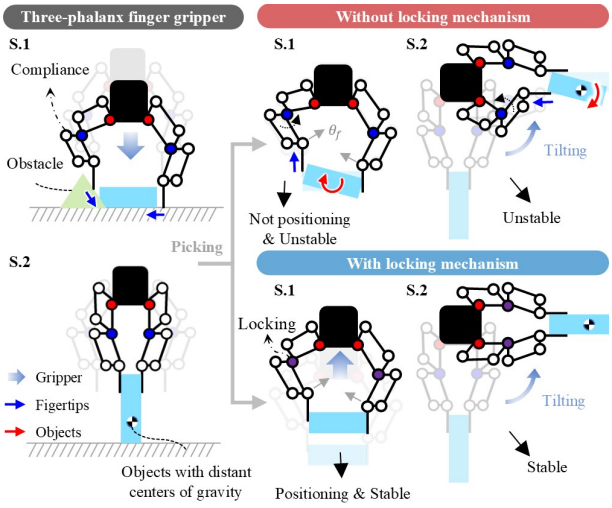


Fig. 3. Describe the behavior of the gripper joints during a pinch grip. The first situation is where an obstacle is placed right next to the target object, and the gripper slides across the surface of the obstacle to grab the target object. The second situation is when you are tilting an object whose center of gravity is outside of the gripping point.

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after the object is grasped. This results in an inability to maintain the grip posture due to changes in the fingertip positions. It is evident that when an external lateral force is applied due to the object's inertia, the passive joint rotates, leading to grip instability. However, with the presence of a locking mechanism, all joints remain fixed in a posture that securely holds the object in all two scenarios. As the grip posture is maintained, the object can be held stably, enabling additional tasks such as stowing or precise manipulation.

B. Locking Principle

Since the locking mechanism must operate passively and only when an object is grasped, a passive joint with a spring is configured at the fingertip to engage only during object contact. As shown in Fig. 4, two types of springs can be used: torsion springs and compression springs. Using a torsion spring alone may result in misalignment of the fingertips due to variations in contact angles, causing the fingertips on both sides to not remain parallel. Even with multiple torsion springs, applying them to the fingertips can complicate the design and potentially lead to instability due to the numerous passive joints. Conversely, a structure with a compression spring can allow for link displacement while maintaining fingertip parallelism. In a pinch grip situation, the compression spring compresses under the grip force, causing a displacement Δd_f. The locking principle based on displacement through spring compression is illustrated in Fig. 5. If, in the absence of slip, the contact point located inside the parallelogram structure at the middle phalanx of the finger, touches the parallelogram before attaining the maximum displacement Δd_f, the finger's movement ceases, thereby fixing its posture. However, if slip occurs after contact, it is noteworthy that in certain postures, the fingertip's force vector θ_f may pull it toward the palm, or the pose of the finger may not be fixed when lifting the object after reacting to the environment. Therefore, a method to achieve locking regardless of θ_f is required, such as by increasing the friction force at the contact point. The unlocking principle relies on the stored energy of the compression spring; when the object is released, this energy returns the mechanism to its original state.

C. Mobility

Mobility analysis assesses the link configuration of the fingertip structure according to the passive-locking principle and the connections between fixed points. The finger mechanism requires varying mobility for different pinch grip scenarios, which are categorized into three types: 1) opening and closing without contact, 2) contact due to environmental factors, and 3) pinch grip situations. The mobility analysis focuses on the links that are not actively driven, as shown in Fig. 6. A link connected to a passive joint is treated as rigid rather than separate if the spring is not engaged. The mobility of the multi-joint finger mechanism is determined using the $Kutzbach\ criterion$ as expressed in (1) [17].

$$m = 3(n-1) - 2j_1 - j_2 \tag{1}$$

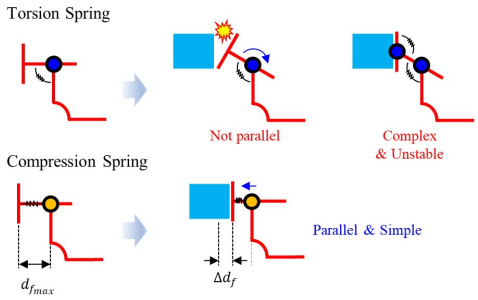


Fig. 4. The type of spring applied to the fingertip

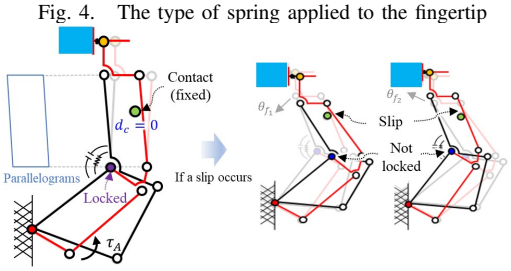


Fig. 5. Passive-Locking principle

Here, j_1 represents the number of joints with 1 degree of freedom, j_2 denotes the number of joints with 2 degrees of freedom, n is the total number of links, and m signifies the total degrees of freedom related to mobility.

Since the gripper's open/close operation occurs in a non-contact situation, the total degree of freedom of the finger structure must be $m = 1$, as it is constrained by the input from a single motor. When external forces are applied, the system incorporates these forces as additional inputs. Consequently, with $m = 2$, the system can effectively respond to external forces by using two inputs to constrain its motion. In a pinch grip scenario, achieving $m = 1$ is desirable for stabilizing the grip while grasping the target object. This configuration ensures that the system responds only to motor inputs, rather than reacting to movements induced by the fingertips' force vectors or unexpected external forces. As shown in Fig. 6, the five-bar linkage, a conventional structure for three-phalanx fingers, encountered issues with degrees of freedom in pinch grip scenarios. The six-bar and seven-bar linkages were found insufficient for maintaining grip stability under normal conditions. The eight-bar linkage was confirmed as the minimum link configuration that satisfies the mobility requirements of the finger mechanism in all three situations depicted in Fig. 7. Table 1 presents the results of mobility analysis for linkage mechanisms ranging from five to eight bars.

III. Analysis of a Passive-Locking Gripper

A. Workspace

The joint angle of the driving link and the passive joint angle relative to the fingertip position can be derived using kinematic analysis of the four-bar linkage based on Plücker coordinates [12,18]. The notation $\hat{S} \in \mathfrak{R}^3$ and $\hat{s} \in \mathfrak{R}^3$ denote the axis and ray coordinates in plane, respectively. This is due to the kinematics of the proposed eight-bar

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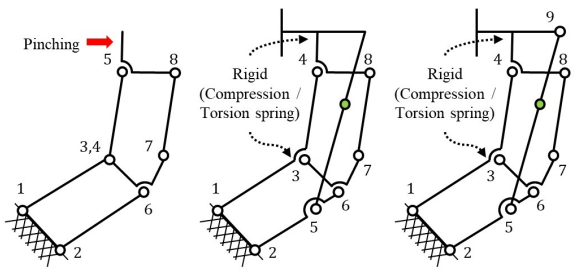


Fig. 6. Analysis of linkage mobility. (a) five-bar linkage in pinch grip situation (b) six-bar linkage in general situation (c) seven-bar linkage in general situation

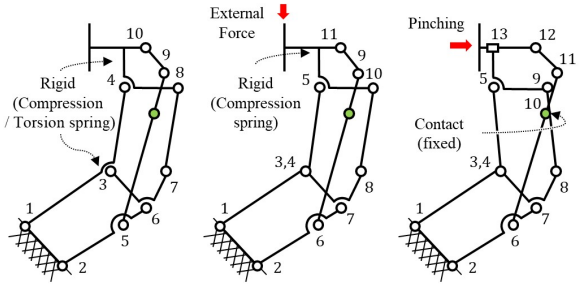


Fig. 7. Analysis of eight-bar linkage mobility in situations. (a) in general situation (b) in external force situation (c) in pinch grip situation

TABLE I

RESULTS OF MOBILITY ANALYSIS BASED ON NUMBER OF LINKS

Linkage	Situation	j_1	j_2	n	m
Five-bar	General	7	0	6	1
	External Force	8	0	7	2
	Pinch grip	8	0	7	2
Six-bar	General	8	0	6	-1
Seven-bar	General	9	0	7	0
Eight-bar	General	10	0	8	1
	External Force	11	0	9	2
	Pinch grip	13	0	10	1

linkage, where the distance d_c between the contact point $\hat{S}_Q$ and the link l_{HI} depends on the compression spring displacement d_f. As illustrated in Fig. 8, the regions where the fingers' actual operation is facilitated can be analyzed by determining the length of d_c when d_f is at its maximum throughout the workspace. When d_f is consistently at its maximum, it represents a typical scenario where the fingertip is not pressed. The value of d_c is calculated based on the coordinates of point $\hat{S}_E$ and the lengths of each link, as derived in (2).

$$d_c = \hat{s}_{HI}^T \hat{S}_Q \tag{2}$$

Fig. 9(a) illustrates the regions where actual operation is feasible by analyzing the collision areas between links and identifying where d_c is negative throughout the entire operational area. It is crucial to consider additional link collision areas during prototype manufacturing, taking into account the thickness of the links. This analysis ensures that all potential collision issues are addressed, accounting for the physical dimensions of the links.

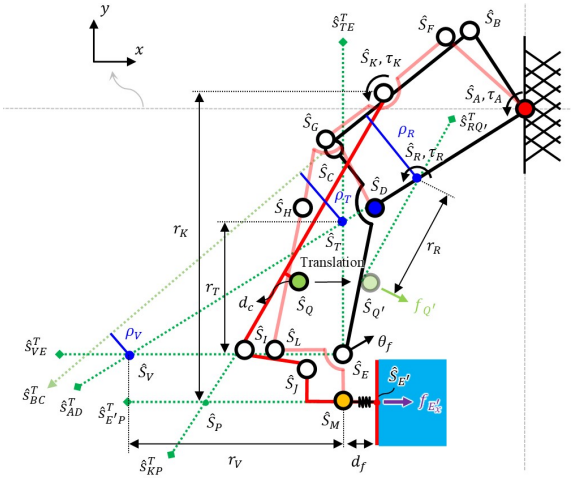


Fig. 8. Schematic structures of the four-bar driven slider-crank and slider movements.

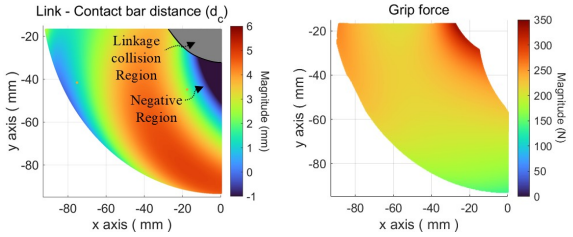


Fig. 9. Analysis of fingertip position: (a) Finger workspace and (b) Grip force in the positive region

B. Grip force

The force vector at the fingertip of a four-bar linkage is a crucial element in the passive-locking principle. The force vector at the fingertip of the four-bar linkage is described by (3).

$$f_{E_x} = \frac{\hat{s}_{BC}^T \hat{S}_T}{\hat{s}_{BC}^T \hat{S}_A} \frac{\tau_A}{r_T} \; for \; \varphi_E = 0°,$$

$$f_{E_y} = \frac{\hat{s}_{BC}^T \hat{S}_V}{\hat{s}_{BC}^T \hat{S}_A} \frac{\tau_A}{r_V} \; for \; \varphi_E = 90°,$$

$$\theta_f = \arctan(f_{E_y}/f_{E_x}) \tag{3}$$

The magnitude of the fingertip force, $f_{E_x'}$, generated by the passive-locking mechanism must be interpreted differently from the conventional four-bar mechanism because the direct application of the input torque, τ_A, originates from $\hat{S}_Q$. Since l_{HI} of the eight-bar linkage acts together with $l_D E$, it can be explained by motion of $\hat{S}_Q$ to $\hat{S}_{Q'}$ as shown in Fig. 8. The contact slope $\varphi_{Q'}$ at the operating point $\hat{S}_{Q'}$ is equal to that of $\hat{s}_{KP}^T$. Therefore, $\hat{S}_R$ is the point of intersection with $\hat{s}_{AD}^T$ where a straight line with slope $\varphi_{Q'}$ passes through $\hat{S}_{Q'}$. We obtain $f_{Q'}$ determining τ_R through the distance from $\hat{S}_R$ to

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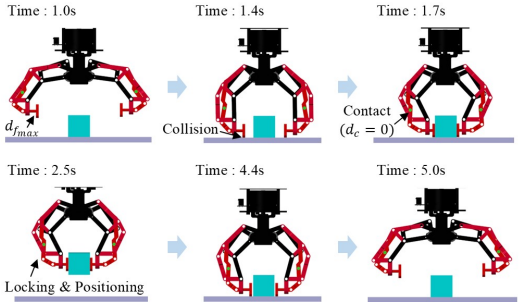


Fig. 10. Pick-and-Place Operations Simulation in MATLAB

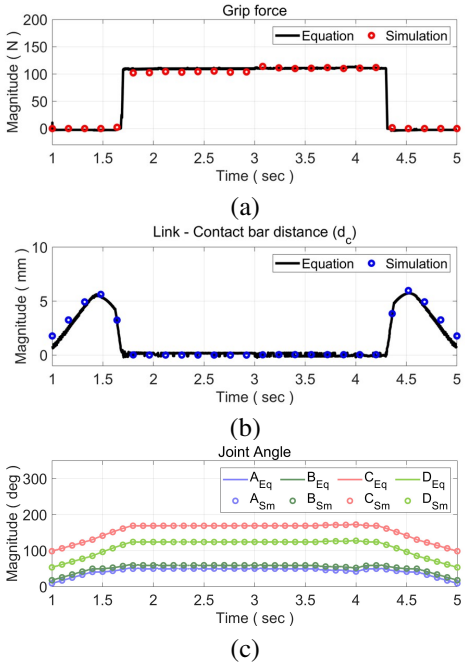


Fig. 11. Graph result to equation and simulation. (a) $f_{E'_x}$ (b) d_c (c) joints angle of $\hat{S}_A$,$\hat{S}_B$,$\hat{S}_C$,$\hat{S}_D$

$\hat{s}^T_{BC}$ and dividing it by r_R, which is the distance between $\hat{S}_R$ and $\hat{S}_{Q'}$ on the straight line $\hat{s}^T_{RQ'}$, $f_{Q'}$ is then multiplied by the distance between $\hat{S}_Q$ and $\hat{S}_K$ on the straight line $\hat{s}^T_{KP}$ to obtain τ_K. Finally, τ_K is divided by r_K, the distance along the y-axis between points $\hat{S}_K$ and $\hat{S}_{E'}$, to determine $f_{E'_x}$, as expressed in (4).

$$f_{E'_x} = \frac{\tau_K}{r_K} = \frac{\hat{s}^T_{BC'}\hat{S}_D}{\hat{s}^T_{BC'}\hat{S}_A}\frac{l_{KQ}\tau_A}{r_K r_D} \tag{4}$$

Fig. 9(b) illustrates the distribution of grip force $f_{E'_x}$ in the region where actual movement is possible. The minimum grip force across the entire area was 72.87 N, and it was observed that greater force could be applied as the grip approached the palm of the gripper.

C. Simulation

A simulation environment was created to validate the functions of the equations derived from analyzing the passive-locking gripper. This environment was developed using MATLAB Simscape Multibody. In the simulation environment, the static friction coefficient between the object and

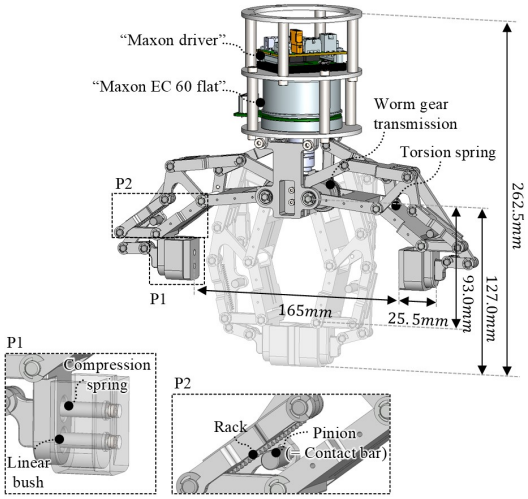


Fig. 12. Prototype model of the proposed gripper design

the gripper is 0.6, the dynamic friction coefficient is 0.4, the collision stiffness between objects is 30,000 N/m, and the damping is 500 N/(m/s). The target object has a width of 40 mm and a mass of 1.5 kg. The simulation results confirmed that the pinch grip motion can adapt to the environment and successfully grasp an object placed on a flat floor, as depicted in Fig. 10. Fig. 11(a) shows that with a constant input torque of 0.420 Nm, the grip can maintain a force of approximately 110 N on the object. Figs. 11(b-c) demonstrate that the distance d_c between the link and the contact point remains fixed at 0 while holding the object, which also keeps the angles of all joints constant. The graph results validated that each equation corresponded accurately with the simulation outcomes.

IV. Experimental Verification

A. Prototype

As shown in Fig. 12, the gripper structure includes one motor, a worm gear with minimal backlash for power amplification and reverse rotation prevention, a torsion spring applied to a four-bar passive joint, a compression spring near the fingertip, and a mechanical stopper that limits the rotation angle of the four-bar mechanism. The motor is MAXON-EC 60 flat with a rated torque of $0.437Nm$, and the gear ratio of the worm gear is $50:1$. The spring consists of two torsion springs with a spring constant of $0.77Nmm/deg$ and two compression springs with a spring constant of $0.98N/mm$ not to interfere with power transmission. In this study, the pinion and rack mechanism prevents slippage between the contact bar and parallelogram. Additionally, two linear bushings are mounted on the fingertip to enhance torsional stability compared to a configuration with a single axis.

B. Experiment Setup

Experiments are conducted in real environments using prototype models to verify the proposed gripper's ability to handle environmental constraints and its passive-locking function. The experiments are carried out with and without

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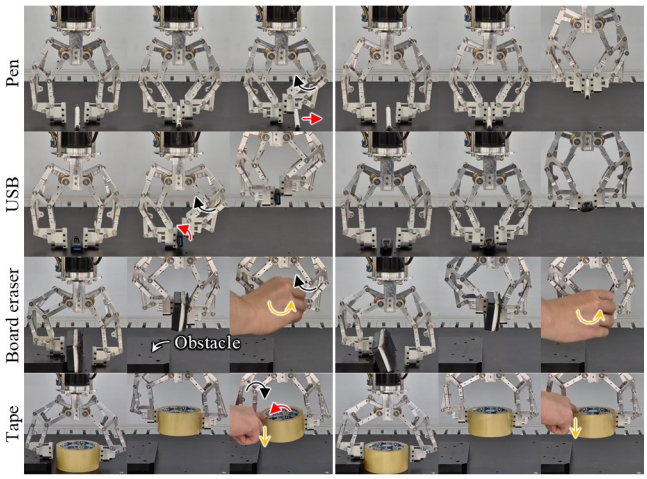


Fig. 13. Picking motion experiments. Respond to environmental constraints: Pen, USB, Board eraser, Tape, respectively.

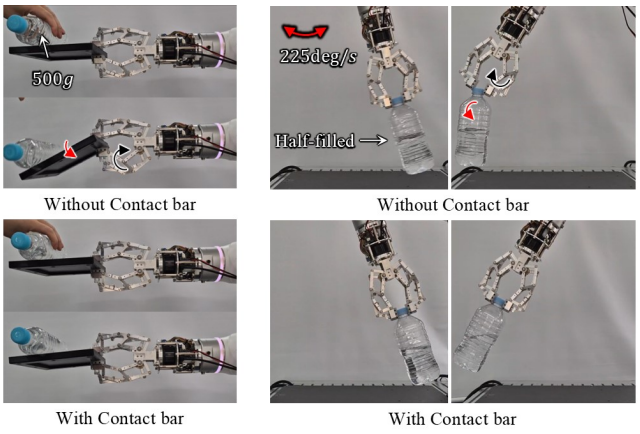


Fig. 14. Robot arm motion and external force application experiments to validate stable grip with and without contact bar. (a) applying 50N external force in lateral direction, (b) grasping a half-filled bottle and swing it from sideways, respectively

the contact bar, a key component of the proposed mechanism, to compare the responses and assess the gripper's locking performance. The surface material at the end is silicone, chosen to enhance friction with the object.

C. Experiment with environmental constraints

The experimental results presented in Fig. 13 demonstrate that, in the absence of the contact bar, the fingertip force vector of the gripper leads to failures in grasping thin or curved objects such as pens and USBs, often resulting in missed grasps or reorientation of the objects. In contrast, objects such as blackboard erasers or tape positioned near an obstacle were effectively grasped despite height discrepancies between the gripper fingers caused by the obstacle. Notably, all objects were successfully grasped irrespective of the presence of the contact bar. However, when external forces were arbitrarily applied after the objects were grasped, changes in the gripper's finger posture altered the orientation and position of the objects. Conversely, when the contact bar was used the orientation and position of the object remained fixed, and the grip maintained stability in all scenarios. This was attributed to the contact bar's ability to mitigate the influence of the terminal force vector after the object had been grasped, thereby ensuring consistent positional stability and secure retention of the object throughout the manipulation process.

D. Experiment with object swing and external forces

The experiment was designed to assess the stability of the grip under unexpected external forces or movements of the robot arm after an object was grabbed, as shown in Fig. 14. The experimental method involved grabbing objects whose center of gravity lay outside the gripping point and swing the gripper sideways or applying an external force while holding a light object, to subject the gripper finger to impacts in both vertical and lateral directions. Fig. 14(a) presents a comparative experiment with a $500g$ object placed while holding a thin, light plate horizontally. The experimental results show that in the absence of the contact bar, the passive joint rotates, causing the gripper to drop the object. In contrast, with the contact bar, the grip posture is fixed, and the object is stably held. In Fig. 14(b), the target object is a half-filled bottle, which subjects the gripper fingers to lateral inertial forces. In the absence of the contact bar, the grip posture is not maintained and objects are dropped when the robot arm is shaken sideways at $225deg/s$ while holding the object. However, when the contact bar is used, the grip posture remains stable and the object is not dropped.

V. CONCLUSION

Under-actuated grippers often struggle with environmental constraints and achieving a stable grip, depending on the number of finger joints. This paper proposes a passive-locking mechanism designed to address these issues. The mechanism responds to environmental constraints before contact with the target object and operates as a rigid one-link finger gripper after contact to ensure a stable grip using a three-phalanx finger structure. Experiments with four different object types demonstrated that the proposed mechanism effectively handles obstacles. To verify its functionality, a swing experiment with a half-full water bottle and a test applying an external force of approximately $5N$ in the lateral direction were conducted with and without the mechanism. The results confirmed that the proposed mechanism overcomes the instability issues typically associated with under-actuated grippers with three-phalanx fingers. It was theoretically possible to maintain a grip even under extreme vertical and lateral forces within its endurance. However, from a design perspective, the number of links and joints exceeds that of existing three-phalanx fingers, and the overall working space of the fingertips decreased by 5.23% due to the constraint of the contact bar's working space. Future work will focus on developing structures with similar functionality but fewer links and joints and optimizing link parameters through algorithm-based design.

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